

Forest Loss and Protected Area Effectiveness for Pollinators (Order: Hymenoptera) in Java-Bali (2000-2024)

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I. INTRODUCTION

Pollinators play an essential role in maintaining ecosystem functions. Among pollinating insects, the order Hymenoptera (bees, wasps, and ants) represent one of the most important groups due to their diverse ecological roles as flower visitors and pollination agents. However, habitat loss caused by deforestation and land conversion are major threats to pollinator communities (Potts *et al.*, 2010). Forests provide important resources for pollinators, such as shelter, food resources, and nesting sites.

Java and Bali represent one of the most modified landscapes in Indonesia, where extensive land conversion has reduced natural forest to agricultural and urban areas. Evaluating pollinator occurrences across forest and non forest landscapes can provide insight into how habitat modification influences pollinator communities in Java and Bali.

Protected areas are established to conserve biodiversity. But while protected areas may preserve suitable habitats, pollinators are widely known to utilize modified landscapes such as urban green spaces and gardens. Therefore, this independent study evaluates the effectiveness of protected areas in Java-Bali in supporting pollinators.

Objectives:

- 1.1 To assess whether forest loss is associated with changes in pollinator (Hymenoptera) occurrence richness across Java-Bali
- 1.2 To evaluate whether remaining forest areas support higher pollinator (Hymenoptera) taxonomic richness compared to forest-loss areas in Java-Bali
- 1.2 To evaluate the role of protected areas in supporting pollinator (Hymenoptera) diversity in Java-Bali

II. METHODS

2.1 DATA ACQUISITION

Pollinator (order Hymenoptera) records were obtained from GBIF, filtered to observed records. Data of tree cover (tree cover 2000) and forest loss year raster (2001-2024) were obtained from Global Forest Change. Protected area polygons (Java-Bali) were obtained from WDPA.

Occurrence records were cleaned by removing records with missing coordinates and taxonomic identification. After cleaning GBIF records, 223 unique pollinator Hymenoptera taxa were recorded across Java and Bali, and further used for analysis. Since the richness data was obtained from observational GBIF data, the pollinator richness will be referred to as observed richness onward.

2.2 FOREST HABITAT CLASSIFICATION

Tree cover values (%) were extracted for each occurrence point using tree cover threshold of 30%, aligned with the Food and Agriculture Organization of the United Nations (FAO) and Global Forest Watch (GFW)'s definition of forest:

Forest = tree cover $\geq$ 30%

Non Forest = tree cover $<$ 30% (Hansen *et al.*, 2013)

Occurrence records and taxon richness were then compared between habitat types to see if occurrences are associated with forest habitat.

2.3 FOREST LOSS ANALYSIS

Forest cover and loss rasters were analyzed to calculate total forest area in 2000, forest loss area from 2001-2024, and percentage of forest loss. Consequently, occurrence records were compared between remaining forest and forest-loss area.

2.4 RICHNESS ANALYSIS

Taxonomic richness was calculated as the number of taxa recorded within each habitat type. Observed taxon richness was calculated in forest and non forest areas, remaining forest and forest-loss areas, and protected and non-protected areas.

2.5 PROTECTED AREA ANALYSIS

Occurrence points were spatially joined with protected area polygons. Records were classified as Protected Area (PA) and Outside Protected Area (OPA). Biodiversity metrics (taxon richness) were compared between categories to evaluate the contribution of protected areas for pollinator conservation (Watson *et al.*, 2014).

2.6 RAREFACTION

Rarefaction was performed to standardize sampling efforts, because sampling efforts differed between habitat categories. Observed taxon richness were estimated and compared after reducing each category to the smallest sample size (Gotelli *et al.*, 2001; Chao *et al.*, 2012).

2.7 GENERALIZED LINEAR MODEL (GLM) ANALYSIS

Predictors used were tree cover percentage and protected area status, while response variable used was observed pollinator richness per grid cell. Poisson GLM was performed because observed richness represents count data, but overdispersion was detected as a result. To avoid error in data interpretation, the final analysis used a negative binomial GLM (Ver Hoef *et al.*, 2007; Zuur *et al.*, 2009).

III. RESULTS AND DISCUSSION

Forest cover analysis showed that Java-Bali experienced a reduction of approximately 410.859,36 ha of forest between 2001 and 2024, representing an estimated 6.39% loss from the Java-Bali forest in 2000 (Figure 3.1 and Figure 3.2). Remaining forest areas contained substantially higher observed pollinator richness, with 221 Hymenoptera taxa recorded compared to only 34 taxa in forest-loss areas (Figure 3.3)

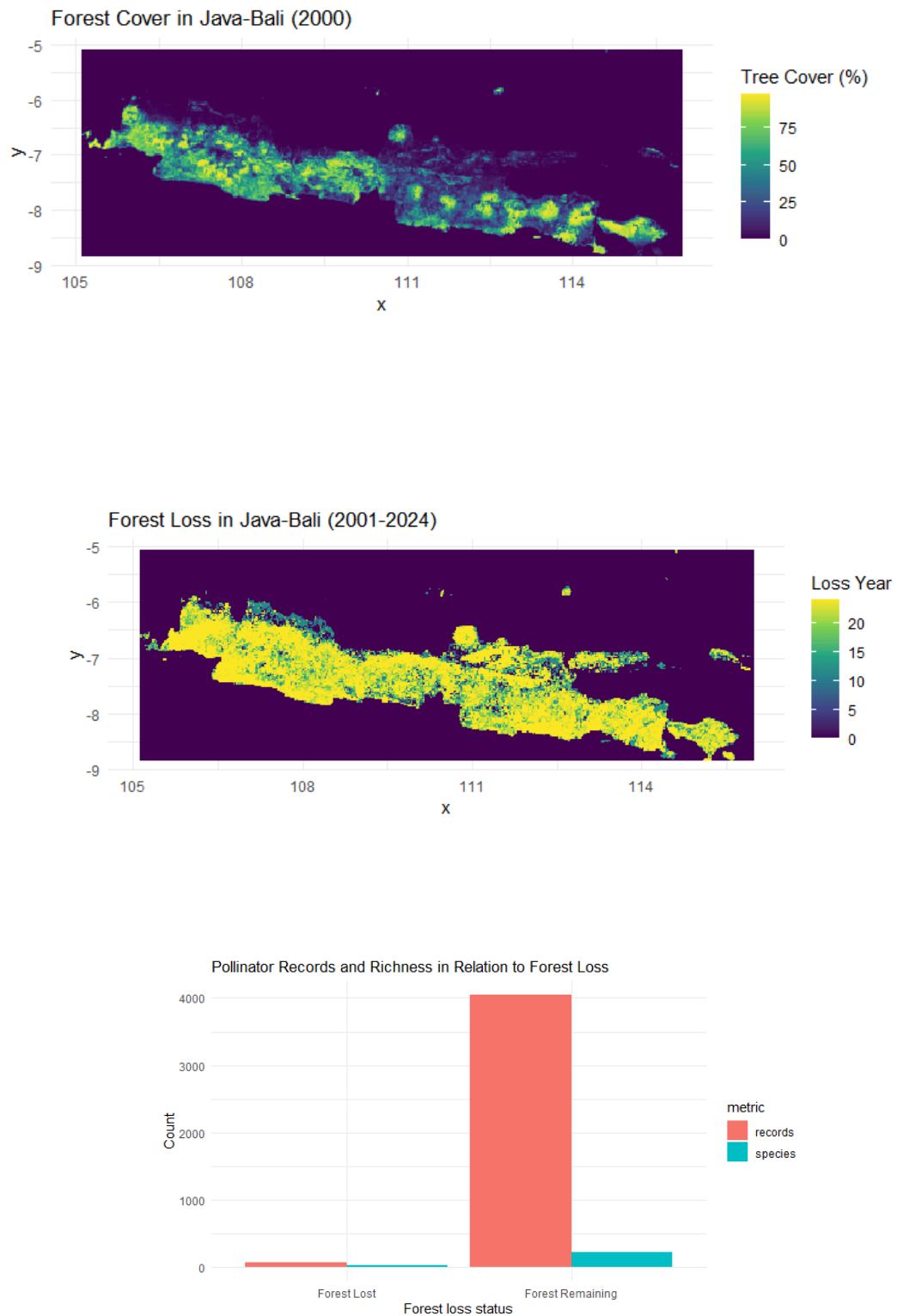


Figure 3.1 Forest cover in Java-Bali in 2000, higher values represent denser forest; **Figure 3.2** Forest loss in Java-Bali over 24 years; **Figure 3.3** Observed pollinator richness in forest-loss areas and remaining forests

The higher richness observed in remaining forest areas may be attributed to the ecological functions of forests. Forests offer diverse floral resources, nesting substrates, microclimatic conditions, and habitat complexity that can support a wide range of Hymenoptera taxa. In contrast, forest-loss areas may experience reduced habitat availability and increased disturbance, which can limit resources for pollinators (Estoque *et al.*, 2019; Potts *et al.*, 2010).

A total of 4.122 Hymenoptera occurrence records were analyzed after data cleaning. Out of these records, 1.348 (32.7%) were located in forest areas, while 2.774 records (67.3%) were located in non forest areas. Observed taxonomic richness showed a similar pattern. A total of 223 unique pollinator taxa were recorded across the study areas. Forest areas contained 146 taxa, while non forest areas contained 180 taxa, representing 34 additional taxa (approximately 23% higher richness) than forest areas (Table 3.1).

Table 3.1 Pollinator occurrence records and observed taxon richness in forest and non forest areas across Java-Bali

Habitat	Observation records	Observed taxon richness
Forest	1.348	146
Non Forest	2.774	180

The spatial distribution of occurrence records indicated that most observations were concentrated in non forest areas, particularly in areas associated with high human activities such as agricultural regions and urban areas. However, these results should not be interpreted as evidence that pollinators prefer non forest areas over forests. Since the occurrence dataset was obtained from GBIF, the observations tend to concentrate in areas that are more accessible, hence creating sampling bias (Meyer *et al.*, 2016). In addition, many Hymenoptera species are capable of utilizing human-modified landscapes, such as gardens, plantations, agroforestry areas, and green spaces that provide abundant floral resources (Baldock *et al.*, 2015). As a result, modified landscapes may contain substantial numbers of pollinator observations despite having lower levels of natural vegetation. Hence, this data does not prove that pollinators prefer non forest areas or deforestation has no effect on pollinators.

Protected Areas (PA) play an important role in conserving pollinator communities by maintaining relatively stable habitats, preserving vegetation, and protecting resources. Global assessments have shown that PA contribute to biodiversity conservation by reducing habitat loss and supporting species persistence, although their effectiveness depends on habitat quality and management practices (Geldmann *et al.*, 2019).

The comparison between Protected Areas (PA) and Non-Protected Areas (OPA) showed that most observed pollinator richness was recorded outside protected areas. In OPA, 3.893 occurrence records representing 216 taxa were recorded, while

PA contained 245 records representing 61 taxa (Figure 3.4). At first glance, this suggests that OPA supported greater observed Hymenoptera richness compared to PA. However, this pattern is strongly influenced by unequal sampling effort. OPA commonly included agricultural areas, gardens, green spaces where surveys are more frequent. Therefore, the higher number of taxa recorded in OPA may partly reflect greater observation intensity rather than higher habitat suitability.

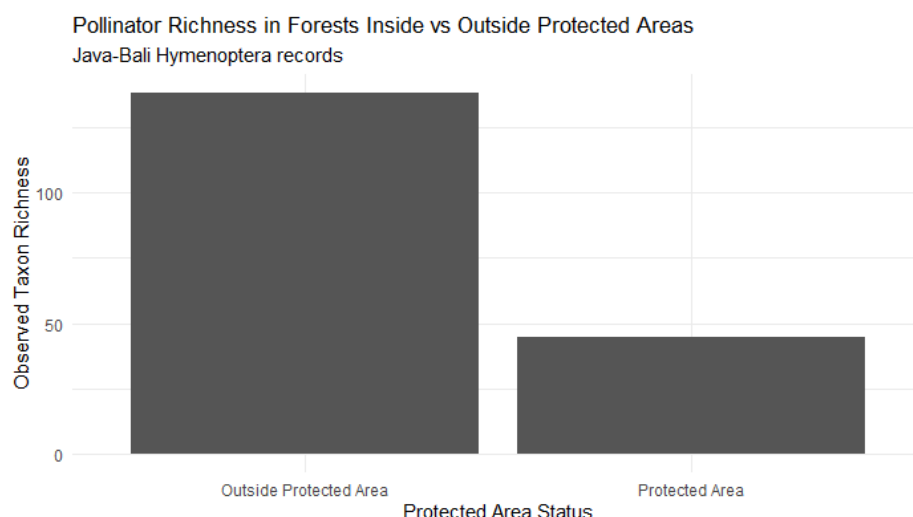


Figure 3.4 Observed pollinator richness in PA and OPA across Java-Bali

To account for this bias, rarefaction was used to standardize richness estimates to an equal sample size of 163 records (Table 3.2). After rarefaction, the difference was reduced with an expected richness of 53.38 taxa in OPA and 45 taxa in PA. This indicates that some of the differences observed in the raw data were driven by sampling intensity rather than true biological differences, and highlights the importance of accounting for uneven survey effort when comparing biodiversity and richness across habitats.

Table 3.2 Comparison of pollinator records and richness in PA and OPA

Category	Occurrence Records	Observed Richness (Taxa)	Rarefied Richness (163 records)
Protected Areas	245	61	45.00
Outside Protected Areas	3.893	216	53.38

**Rarefied richness was standardized to 163 occurrence records to account for unequal sampling effort between categories*

Despite the rarefaction results, the Generalized Linear Model (GLM) showed protected areas were positively associated with observed pollinator richness (Table

3.3). PA supported approximately 23.4% higher expected pollinator richness than OPA (multiplier = 1.234, 95% CI = 1.170-1.301). This suggests that protected areas may still contribute to maintaining pollinator biodiversity and richness. Protected forests likely provide more stable habitat conditions (reduced disturbance, nesting resources) and more diverse plant communities that support a wide range of Hymenoptera taxa (Geldmann *et al.*, 2019).

In contrast, forest cover was negatively associated with observed pollinator richness (multiplier = 0.994, 95% CI = 0.993-0.995). This suggests that with each 1% increase in tree cover, there is a small decrease in expected richness. This finding contrasts with the expectation that dense forest areas would support greater pollinator diversity and richness. Agricultural areas, gardens, and green spaces can provide abundant floral resources that attract more pollinators. As a result, pollinator richness may remain relatively high in areas with lower tree cover. Furthermore, factors such as floral diversity, nesting resources, and vegetation structure can also influence species richness. This statement is supported by Winfree *et al.* (2009) which stated the idea that pollinator responses to disturbances are not always linear, since some taxa tolerate or moderately benefit from disturbed landscapes.

Table 3.3 Association between tree cover, PA, and pollinator richness

Predictor	Multiplier	95% Confidence Interval (CI)
Tree cover	0.994	0.993-0.995
Protected area	1.234	1.170-1.301

**The numbers suggest that for every 1% increase in tree cover, richness decreases by 0.6%. After accounting for tree cover, richness in protected areas is ~23.4% higher*

Protected Areas were found to play an important role in supporting pollinator diversity. Although raw richness data were affected by differences in sampling effort, the negative binomial GLM analysis indicated that PA status was positively associated with pollinator richness. Despite the significance of protected areas, pollinator conservation may require a more detailed approach to utilize landscapes with high human activities. However, areas like gardens and green spaces may be vulnerable to land use conversion and pesticide use.

Future conservation strategies should focus on maintaining and improving habitat quality across both PA and OPA. Within PA, management should prioritize forest preservation, restoration of degraded habitats, and protection of native flowering plants. In OPA, society can implement pollinator-friendly practices such as reducing pesticide usage and promoting biodiversity-friendly agricultural practices to help sustain pollinator populations. Overall, the findings suggest that PA are effective components of pollinator conservation in Java-Bali, but their role should be complemented by broader conservation management that supports pollinators across forests, agricultural, as well as urban areas.

IV. RESEARCH GAPS

4.1 Biodiversity occurrence databases are strongly influenced by uneven sampling effort. Future research should incorporate standardized field surveys across both forest areas and non forest areas to better distinguish ecological patterns from sampling bias.

4.2 This study assessed forest cover as an indicator of habitat availability, but tree cover alone may not fully represent habitat quality for pollinators. Future research should integrate additional environmental variables to identify which habitat characteristics determine pollinator richness and diversity.

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